

ELC 2060

Electronic Circuits

Output Stages and Power Amplifiers

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Engineering

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Introduction

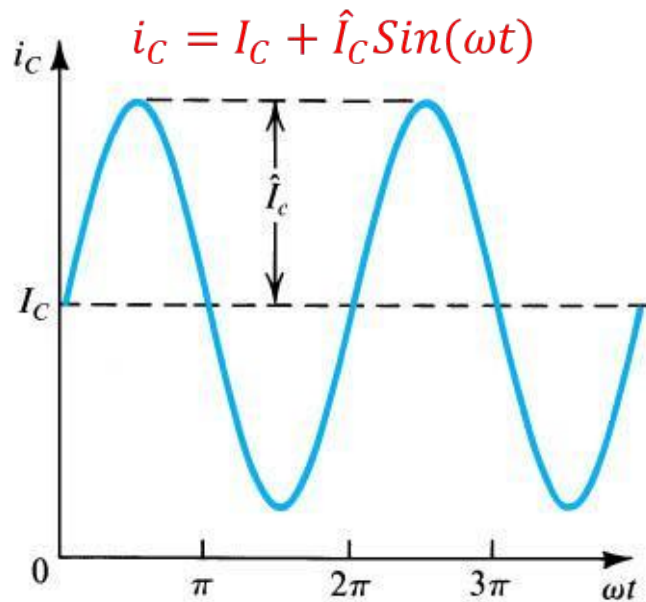
- Output stages that drive small load resistor with high output power should provide
 - Large voltage and current swings High Power
 - Low output impedance to prevent gain loss
- Typical applications
 - Op-amp output stages
 - Power amplifiers for wireless transmitters and audio applications

Introduction

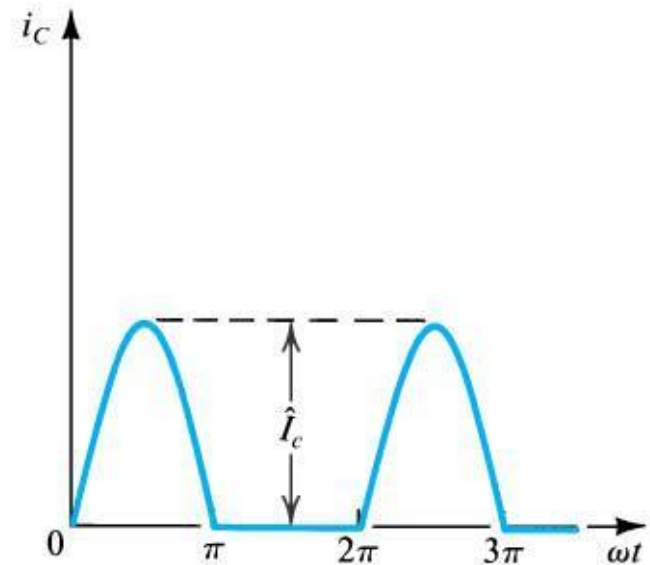
- Output stages deal with large signals
 - Small signal approximation must be used with care
- Efficiency is defined as the ratio between output power to DC power consumed by the amplifier
 - $\eta = P_{\text{out}} / P_{\text{DC}}$
 - $P_{\text{out}} = 1\text{W}$, $\eta = 50\%$, $P_{\text{DC}} = 2\text{W}$
 - $P_{\text{out}} = 1\text{W}$, $\eta = 80\%$, $P_{\text{DC}} = 1.25\text{W}$
 - $P_{\text{out}} = 1\text{W}$, $\eta = 90\%$, $P_{\text{DC}} = 1.11\text{W}$
- P_{DC} & η are very important because they affect the battery lifetime & max junction temperature

Power Amplifier Classes

- Power amplifiers are classified according to their current waveforms
 - Class “A” PA current has 360° conduction angle
 - Class “B” PA current has 180° conduction angle
- Conduction angle, is the period in degrees (or radians) in which the PA main device is conducting current



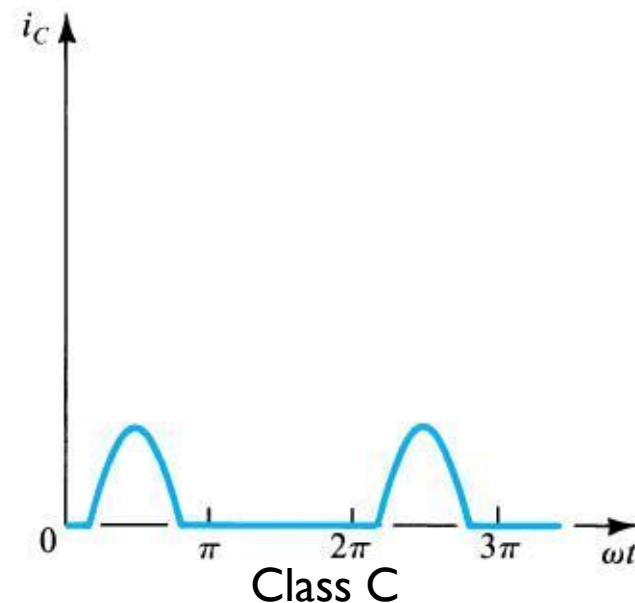
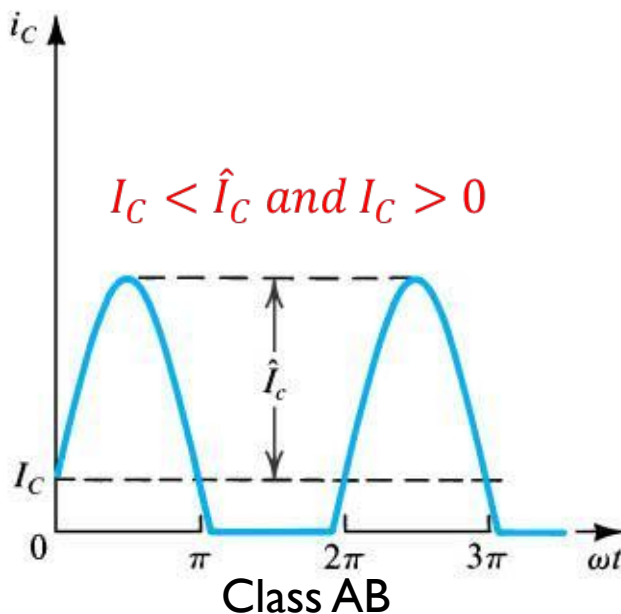
Class A $I_C + \hat{I}_C > 0$



Class B $I_C = 0$

Power Amplifier Classes

- Power amplifier classified according to their current waveforms
 - Class “**AB**” PA conduction angle $>180^\circ$ and $<360^\circ$
 - Class “**C**” PA conduction angle $< 180^\circ$



Emitter Follower as Output Stage

- Emitter follower has low output impedance
- Small signal results

$$G = \frac{g_m R_L}{1 + g_m R_L}$$

$$R_{out} \approx 1/g_m$$

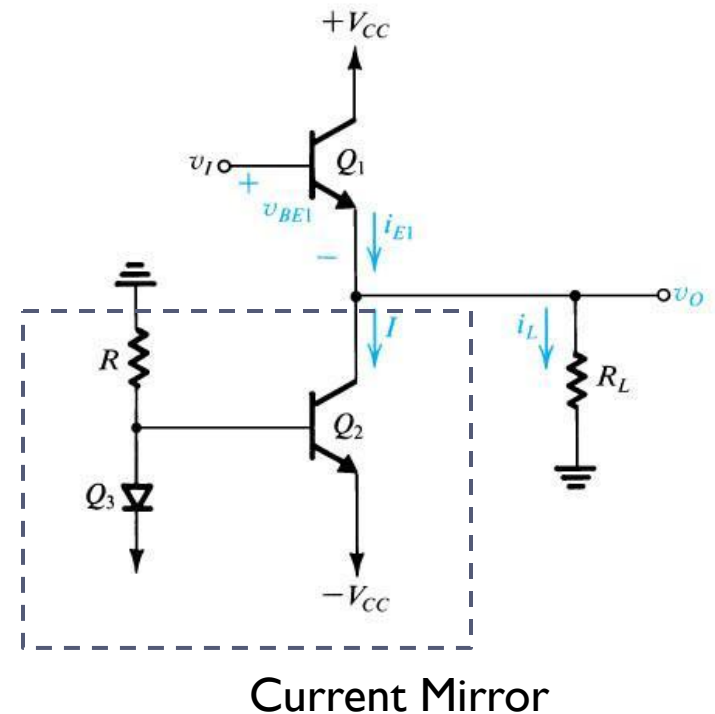
$$R_{in} \approx r_\pi + (\beta + 1)R_L$$

- Large signal analysis:

$$v_O = v_I - v_{BE1}$$

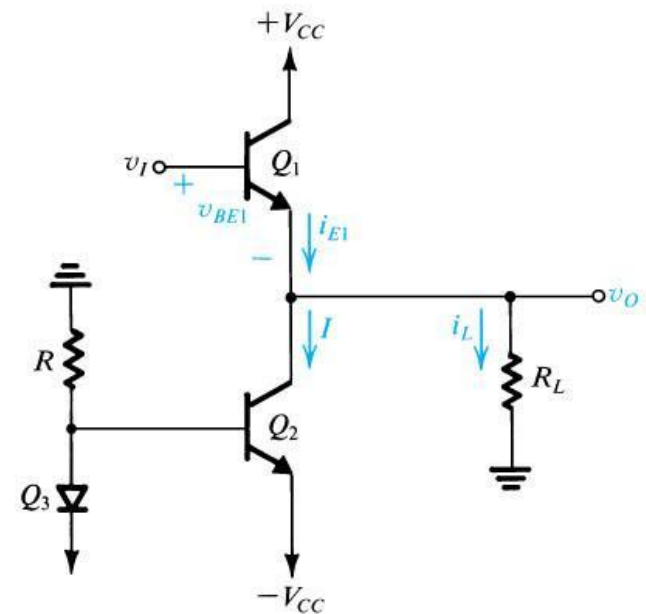
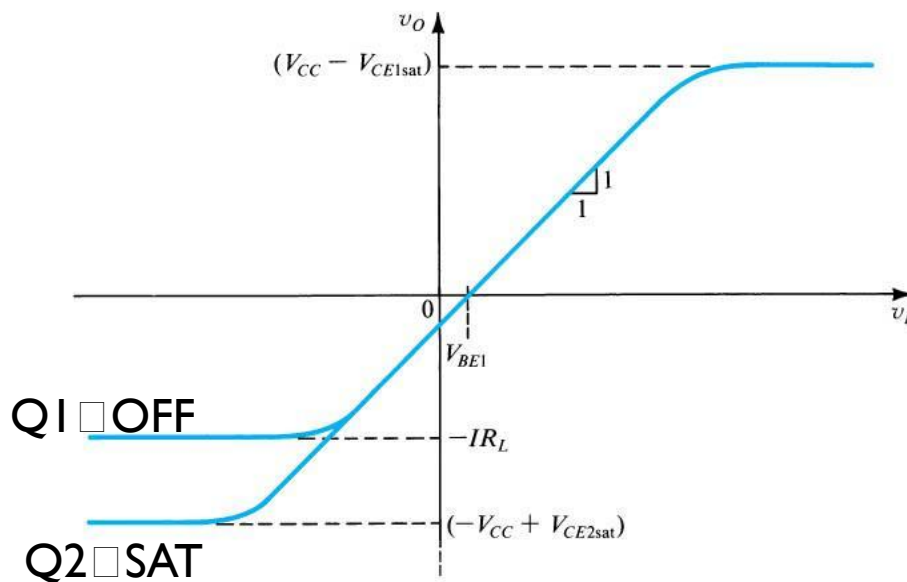
$$i_{E1} = I + i_L$$

- v_{BE1} is a function of the current ($G < 1$)



Emitter Follower as Output Stage

- Output signal follows the input signal
 - Max output signal is $V_{CC} - V_{CE1sat}$
 - Min output signal is
 - $-V_{CC} + V_{CE2sat}$ If Q2 enters saturation first
 - $-IR_L$ if Q1 turns off first ($i_{E1}=0, i_L=-I$)
 - For maximum swing, $I > |-V_{CC} + V_{CE2sat}| / R_L$

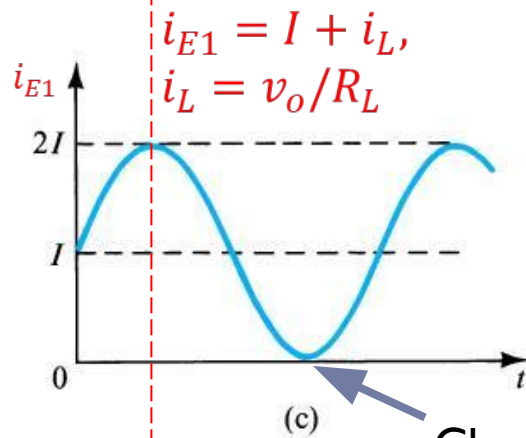
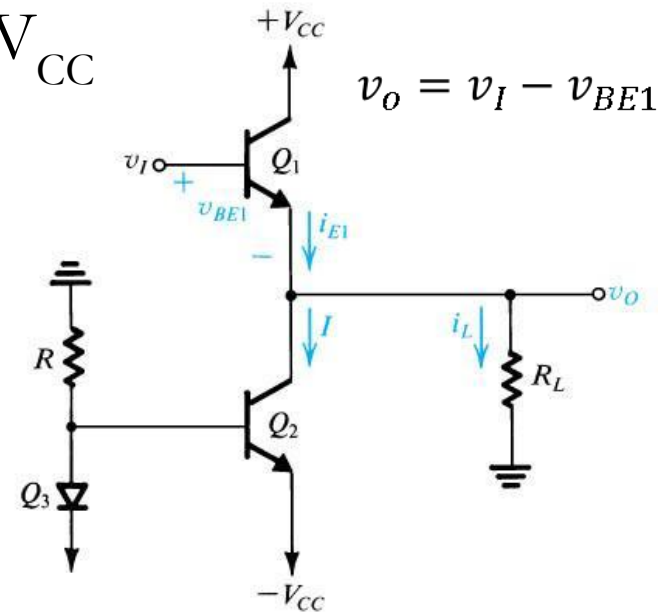
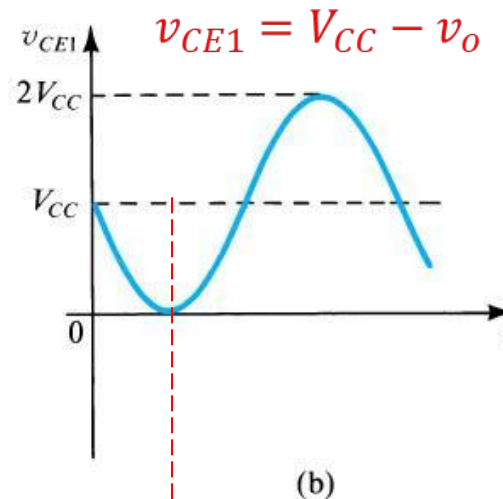
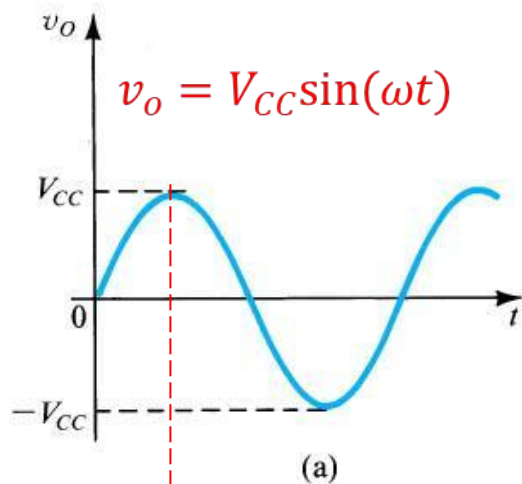


$$v_O = v_I - v_{BE1}$$

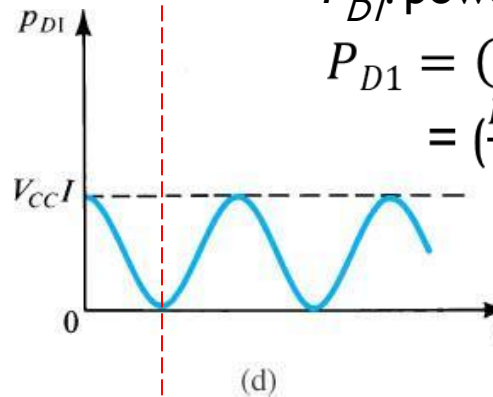
$$i_{E1} = I + i_L$$

Emitter Follower as Output Stage

- Neglecting the effect of V_{CE1sat} , $\max v_o$ is V_{CC}



Class A



P_{D1} : power dissipated in Q1

$$P_{D1} = (I + I \sin \omega t)(V_{CC} - V_{CC} \sin \omega t)$$

$$= \left(\frac{IV_{CC}}{2} + \frac{IV_{CC}}{2} \cos 2\omega t \right)$$

Power conversion efficiency

▣ $\eta = \text{Load power} / \text{Supply power}$

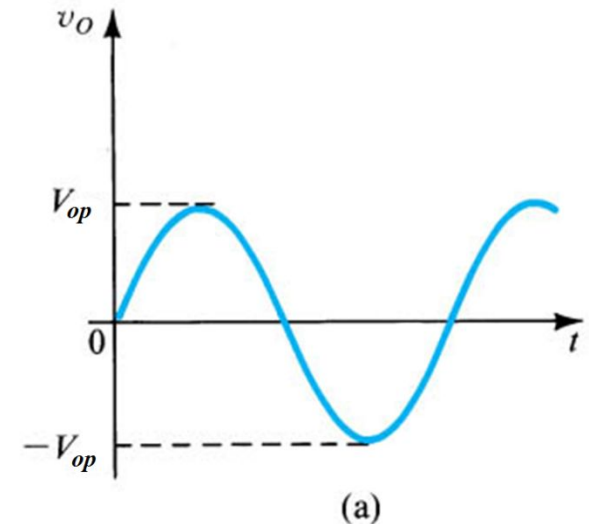
▶ $P_L = V_{op}^2 / 2R_L$

▶ $P_S = 2V_{CC}I$

▶ $\eta = \frac{V_{op}^2}{4I R_L V_{CC}} = \frac{1}{4} \left(\frac{V_{op}}{I R_L} \right) \left(\frac{V_{op}}{V_{CC}} \right)$

▶ Since $V_{op} \leq V_{CC}$ and $V_{op} \leq I R_L$

▶ Maximum class A efficiency = 25 %



Emitter Follower as Output Stage

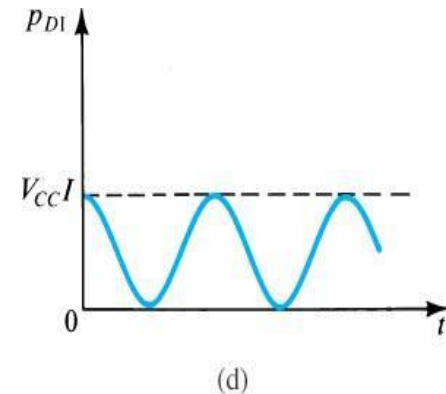
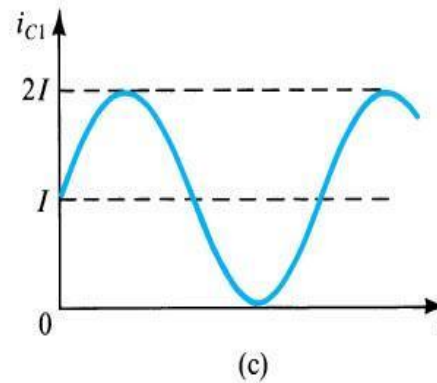
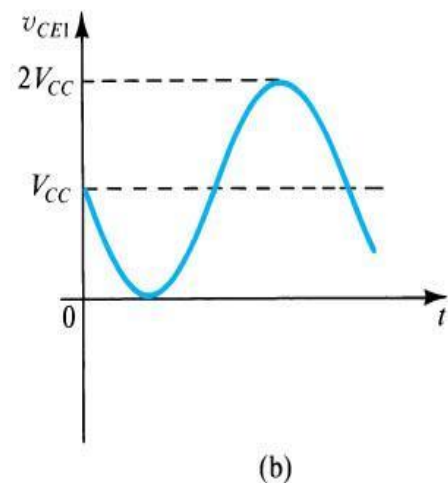
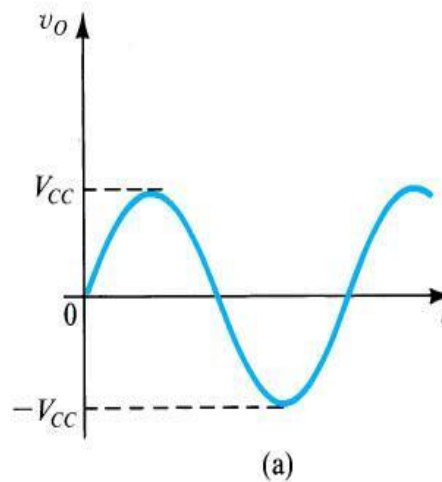
- At max voltage swing and max efficiency

$$V_{op} = V_{CC}, I = V_{CC}/R_L$$

- Power dissipated in $Q_1 = V_{CC}I/2$
- Power dissipated in $Q_2 = V_{CC}I$
- Power dissipated in load $= V_{CC}I/2$
- DC input power is $2V_{CC}I$

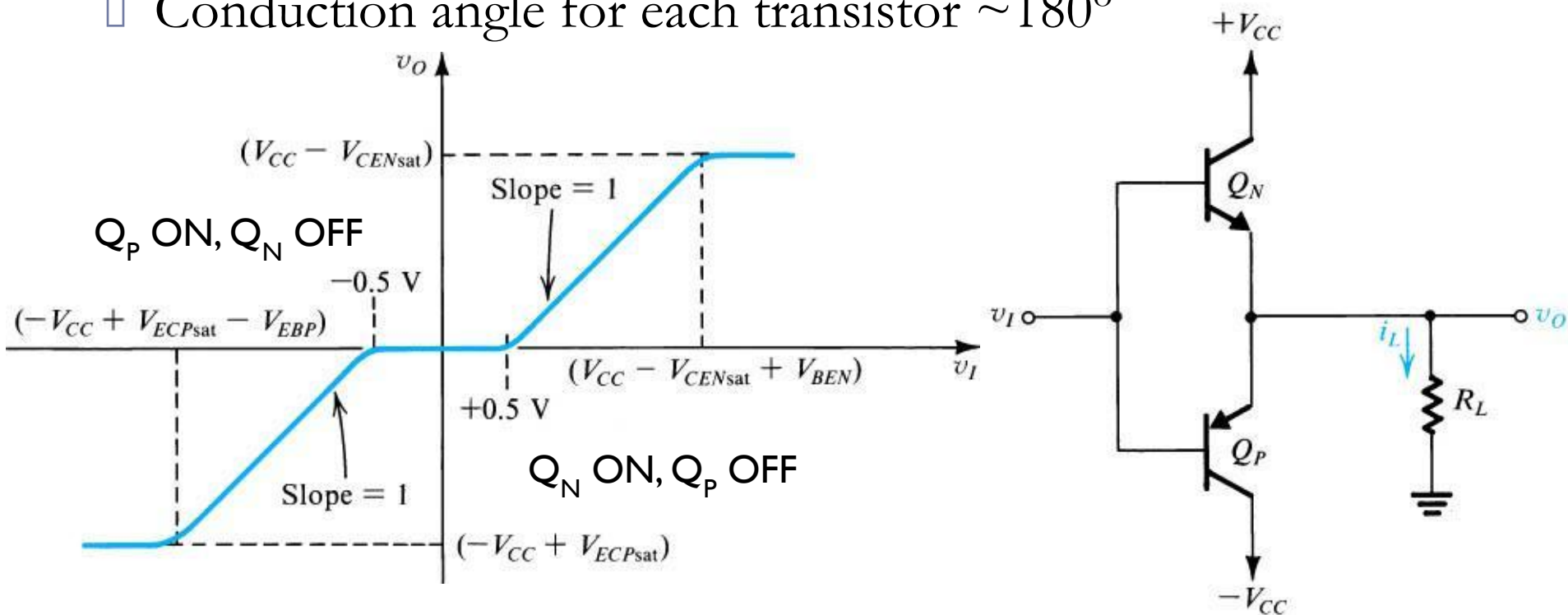
P_{D2} : power dissipated in Q_2

$$P_{D2} = I(V_{CC} + V_{CC}\sin\omega t)$$



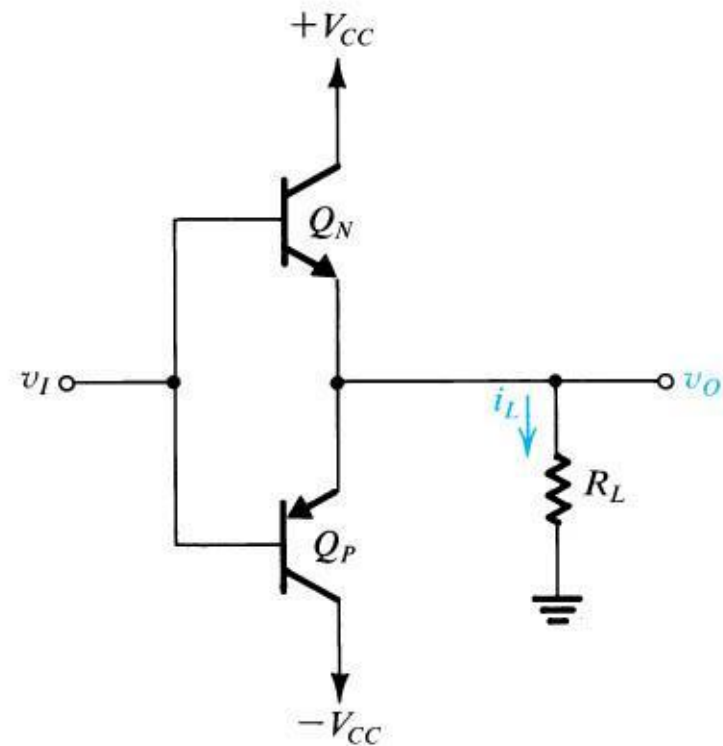
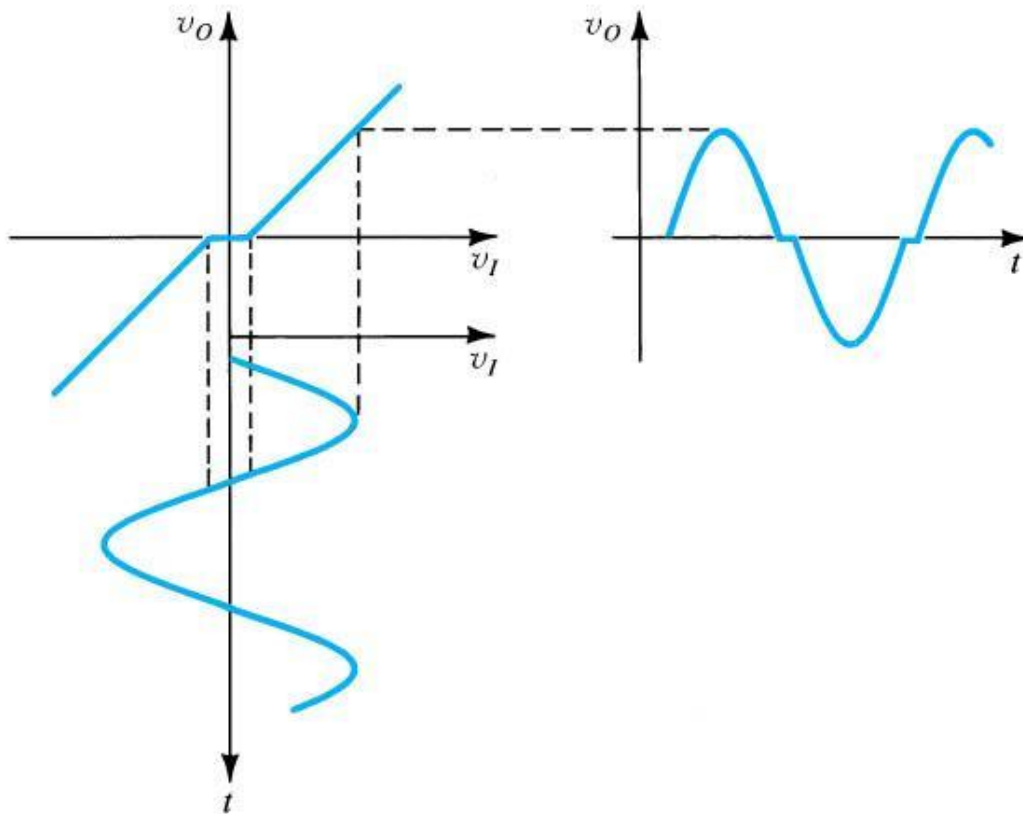
Class B output stage

- Both transistors are off when the input is zero
- Q_N turns on when $v_I > V_{BEN}$, then $v_o = v_I - V_{BEN}$
- Q_P turns on when $v_I < -V_{EBP}$, then $v_o = v_I + V_{EBP}$
- Conduction angle for each transistor $\sim 180^\circ$



Class B output stage

- There is a region where both transistors are off
- Cross over distortion will occur



Efficiency of class B output stage

- ▣ Neglecting the cross over distortion
 - ▶ Load power $P_L = V_{op}^2 / 2R_L$
 - ▶ The supply current is a half sine wave with amplitude $I_{op} = V_{op} / R_L$
 - ▶ Using Fourier series, the DC average current is $I_{av} = V_{op} / \pi R_L$
 - Supply power $P_S = 2I_{av} V_{CC} = 2V_{op} V_{CC} / \pi R_L$
 - ▶ $\eta = \frac{P_L}{P_S} = \frac{\pi}{4} \frac{V_{op}}{V_{CC}}$
 - ▶ Class B $\eta_{max} = \pi/4 = 78\%$ at $V_{op} = V_{CC}$

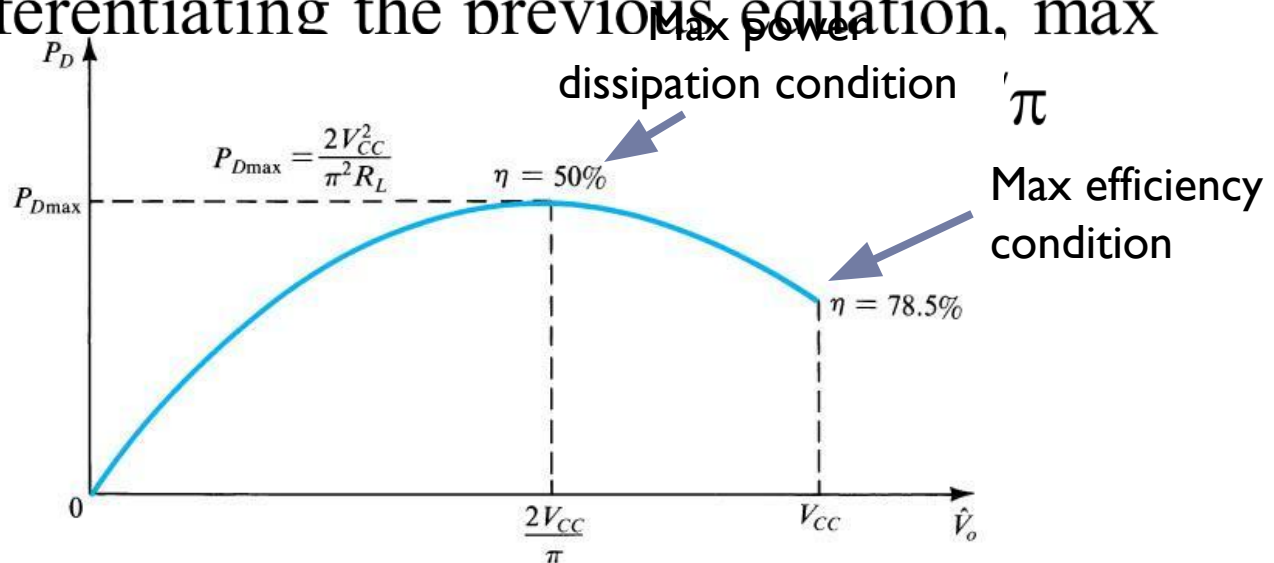
Power dissipation of class B

- Power dissipated in class B stage is given by

- $P_D = P_S - P_L$

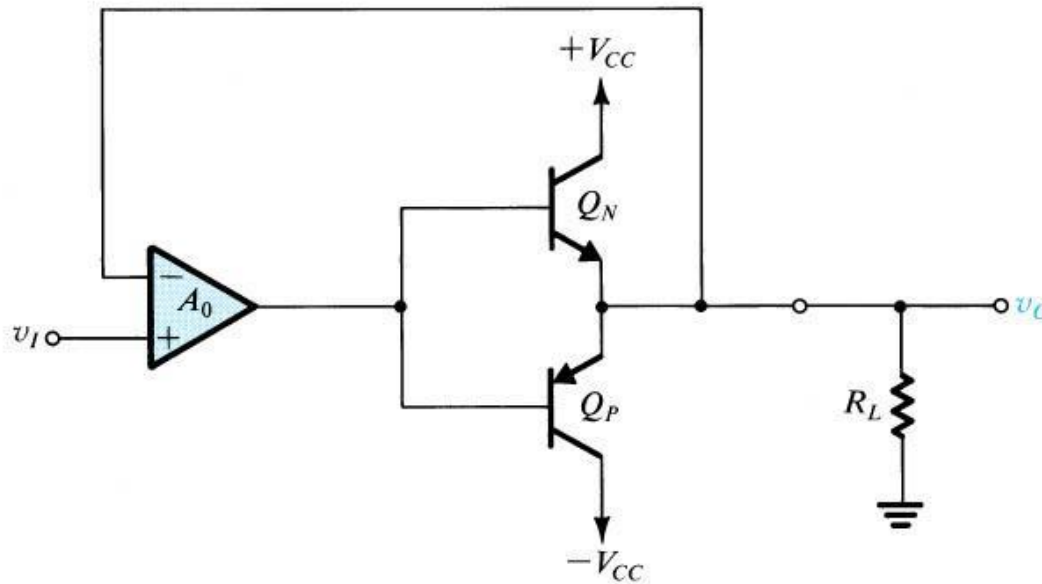
- $P_D = \frac{2}{\pi} \frac{V_{op}}{R_L} V_{CC} - \frac{1}{2} \frac{V_{op}^2}{R_L}$

- By differentiating the previous equation, max power



Reducing cross over distortion

- Dead zone is reduced to $\pm V_{BE}/A_0$



Class AB output stage

❏ Cross over distortion can be eliminated by biasing transistor using a small current

▶ For $V_{in}=0$ then $V_{out}=0$,

▶ $i_N = i_P = I_Q = I_S e^{V_{BB}/2V_T}$

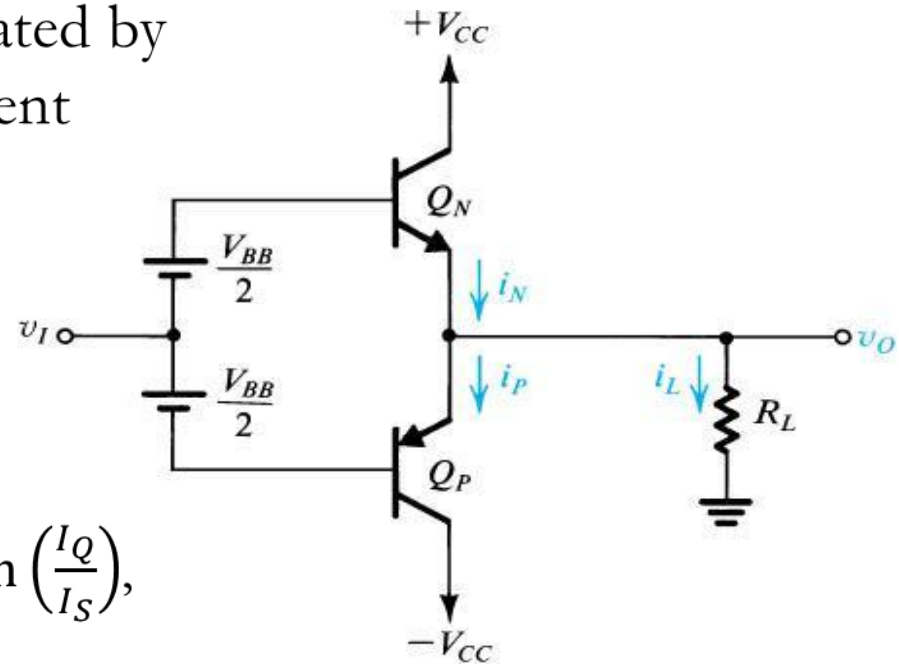
▶ $V_{BEN} + V_{EBP} = V_{BB}$

▶ In general

▶ $V_T \ln\left(\frac{i_N}{I_S}\right) + V_T \ln\left(\frac{i_P}{I_S}\right) = V_{BB} = 2V_T \ln\left(\frac{I_Q}{I_S}\right)$,
then $i_N i_P = I_Q^2$ (1)

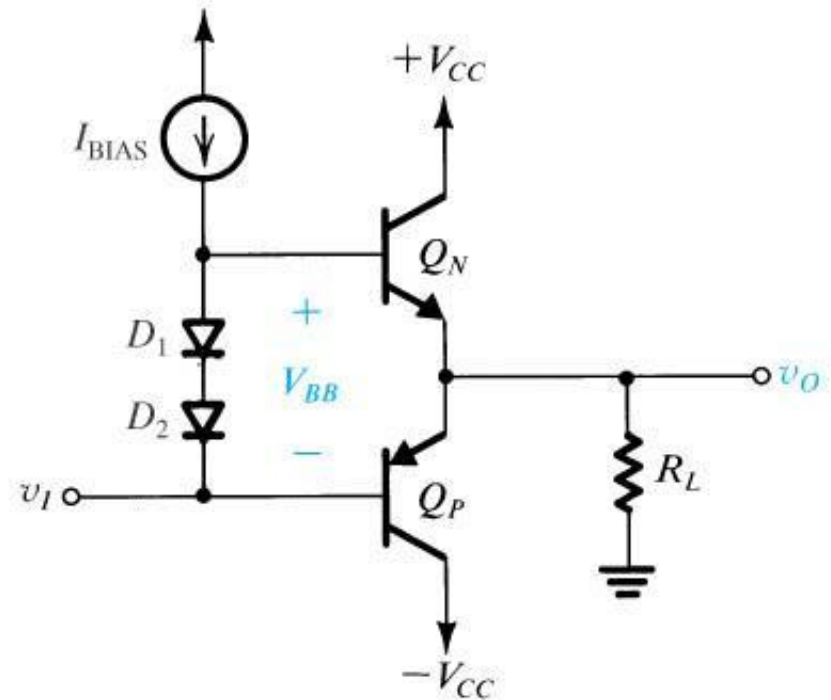
▶ $i_N = i_P + i_L$ (2)

▶ From (1) and (2) $i_N^2 - i_L i_N - I_Q^2 = 0$



Biasing the class AB amplifier

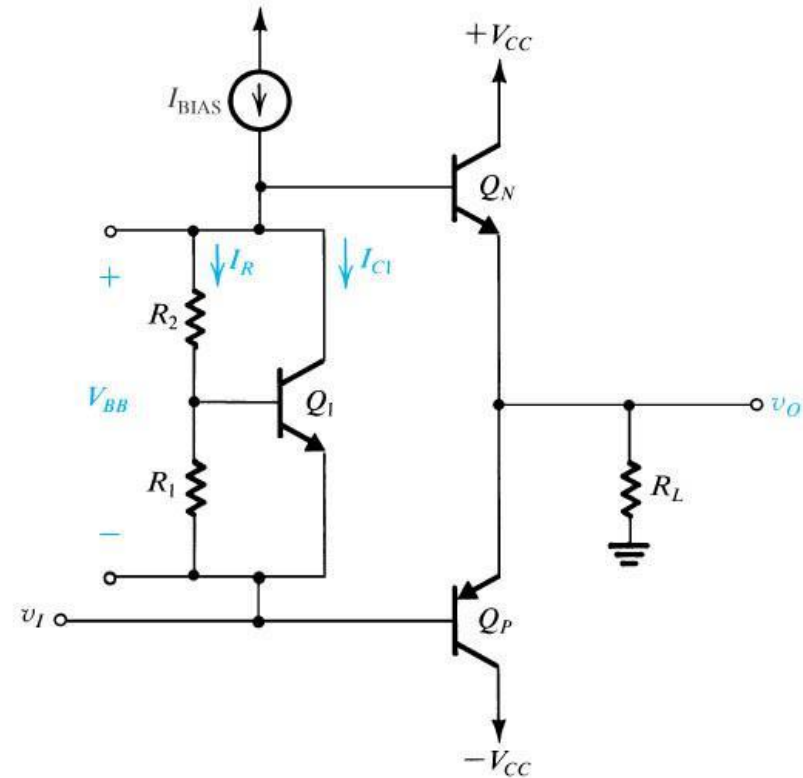
- $I_Q = nI_{BIAS}$ where n is the ratio between the emitter area of BJTs and area of diodes
- Note that I_{BIAS} should be greater than maximum of i_{BN}
 - $I_{BIAS} > i_{D1,2} + i_{BN}$



Biasing the class AB amplifier (V_{BE} multiplier)

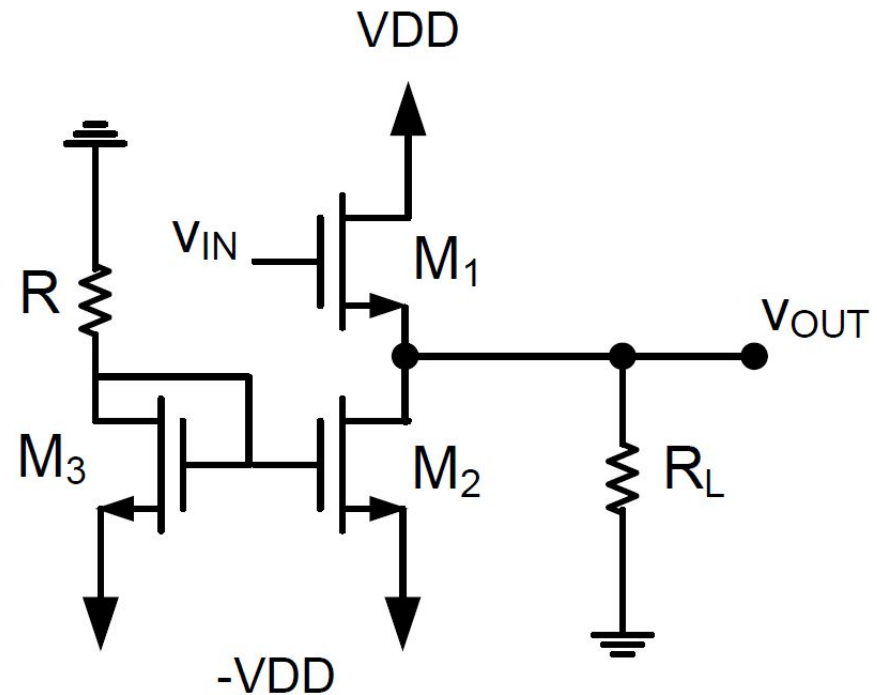
► Neglecting the base current of Q_1

- $I_R = \frac{V_{BE1}}{R_1}$
- $V_{BB} = I_R(R_1 + R_2) = V_{BE1} \left(1 + \frac{R_2}{R_1}\right)$
- $V_{BE1} = V_T \ln \left(\frac{I_{C1}}{I_{S1}}\right) = V_T \ln \left(\frac{I_{BIAS} - I_R}{I_{S1}}\right)$



Using MOS in output stages

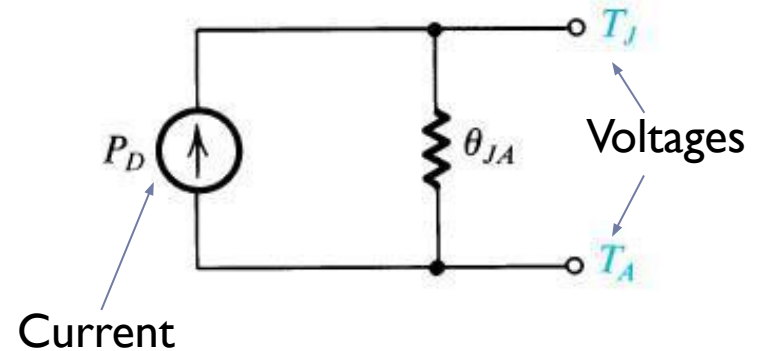
- MOS can be used in output stage instead of bipolar
- Main difference is that V_{GS} of MOS device is a function of current
- At peak output voltage V_{omax}
 - $V_{omax} = V_{DD} - V_{ov}$
 - $V_{ov} = \sqrt{\frac{2(I + V_{omax}/R_L)}{k}}$



Thermal Resistance

- Power transistors dissipate large power converted into heat
- The temperature rise is proportional with the power consumption:

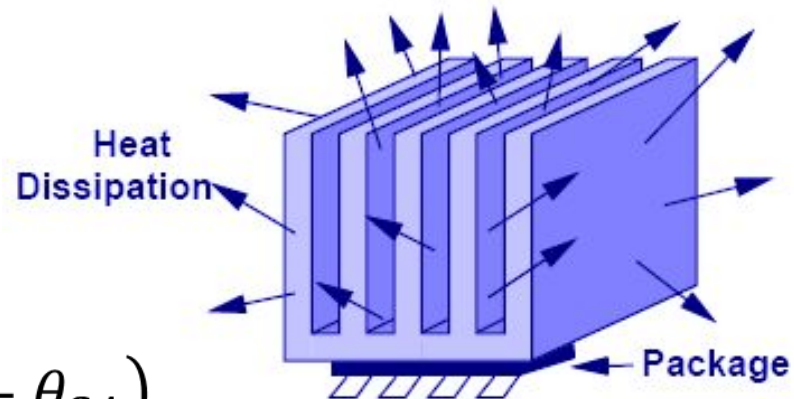
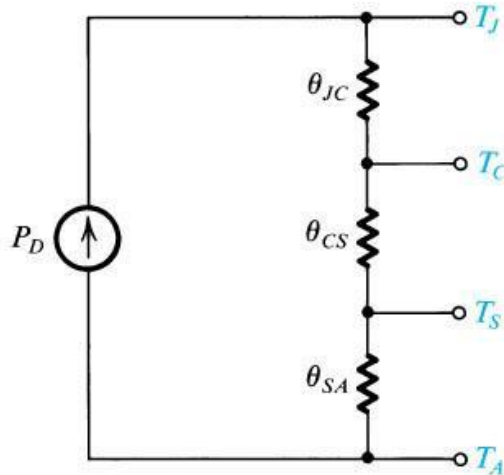
$$T_J - T_A = \theta_{JA} P_D$$



- T_A is the ambient temperature, T_J is the junction temperature
- T_J shouldn't exceed $T_{J_{\max}}$ ($\sim 125^\circ\text{C}$ for Si)
- θ_{JA} is the thermal resistance between junction & ambient ($^\circ\text{C}/\text{W}$)
- Thermal conduction is analogous to electrical conduction

Thermal Resistance (Cont'd)

- ▶ To reduce dissipate the heat out of the circuit a heat sink is used



$$T_J - T_A = P_D(\theta_{JC} + \theta_{CS} + \theta_{SA})$$

- ▶ θ_{JC} : **j**unction to **c**ase thermal resistance
- ▶ θ_{CS} : **c**ase to heat **s**ink thermal resistance
- ▶ θ_{SA} : heat **s**ink to **a**mbient thermal resistance